Cybersecurity Incident Report: Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.
The UDP protocol reveals that: DNS queries sent over UDP from the client (192.51.100.15) to the DNS server (203.0.113.2) are not being delivered. Each attempt to resolve the A record for www.yummyrecipesforme.com (query ID 35084) failed, so the browser never received the IP address it needs to send the HTTPS request to the web server. This is why customers cannot load the website. This is based on the results of the network analysis, which show that the ICMP echo reply returned the error message: “udp port 53 unreachable”. This was returned by the DNS server (203.0.113.2) as an ICMP destination unreachable message. The query was sent three times, and the same error came back every time. The port noted in the error message is used for: DNS (Domain Name System). Port 53 is the standard port for DNS queries, which translate domain names into IP addresses. The most likely issue is: The DNS server is not responding to DNS requests: no service is listening on UDP port 53 at 203.0.113.2. Either the DNS service has stopped or been misconfigured, or traffic to port 53 is being blocked (for example, by a firewall rule). Without DNS resolution, users cannot reach the website, even though the web server itself may be working normally.

Part 2: Explain your analysis of the data and provide at least one cause of the incident.
Time incident occurred: 1:24 p.m. (13:24:32.192571 in the tcpdump log). This is when the analyst reproduced and captured the failure. Customers had reported the problem before this capture was taken. Explain how the IT team became aware of the incident: Several customers of the client reported that they could not access www.yummyrecipesforme.com and saw the error “destination port unreachable” after waiting for the page to load. These reports were passed to the IT team for investigation. Explain the actions taken by the IT department to investigate the incident: The analyst visited the website and reproduced the same “destination port unreachable” error. The analyst then ran the network protocol analyzer tcpdump and loaded the webpage again to capture the traffic. The log was reviewed to trace the outgoing DNS

queries (UDP) and the responses returned (ICMP). The findings were reported to the analyst's direct supervisor. The incident is ongoing and is currently being handled by the security engineers.

Note key findings of the IT department's investigation (i.e., details related to the port affected, DNS server, etc.):

The client (192.51.100.15) sent DNS A-record queries over UDP to the DNS server (203.0.113.2) on port 53. The query ID was 35084, and the "+" indicates flags were set on the request. Each query received an ICMP response from 203.0.113.2 with the error "udp port 53 unreachable". The query was retried twice more, and both retries failed with the same error. Because DNS resolution failed, the browser never obtained an IP address, so no HTTPS request reached the web server. The fault therefore lies with the DNS service on port 53, not with the web server or HTTPS (port 443).

Note a likely cause of the incident:

The DNS service on 203.0.113.2 is not listening on UDP port 53. This could be because the service has stopped, crashed, or been misconfigured, or because a firewall rule is blocking inbound traffic on port 53. A denial-of-service (DoS) attack against the DNS server is also possible and should be ruled out.

Next steps:

1. Check whether the DNS service on 203.0.113.2 is running and listening on port 53, and restart it if it has stopped.
2. Review firewall and access control rules for any recent change that blocks port 53.
3. Review DNS server logs and traffic volumes for signs of a DoS attack or resource exhaustion.
4. Once the service is restored, confirm that the domain resolves and that the website loads. Consider adding a secondary DNS server for redundancy.